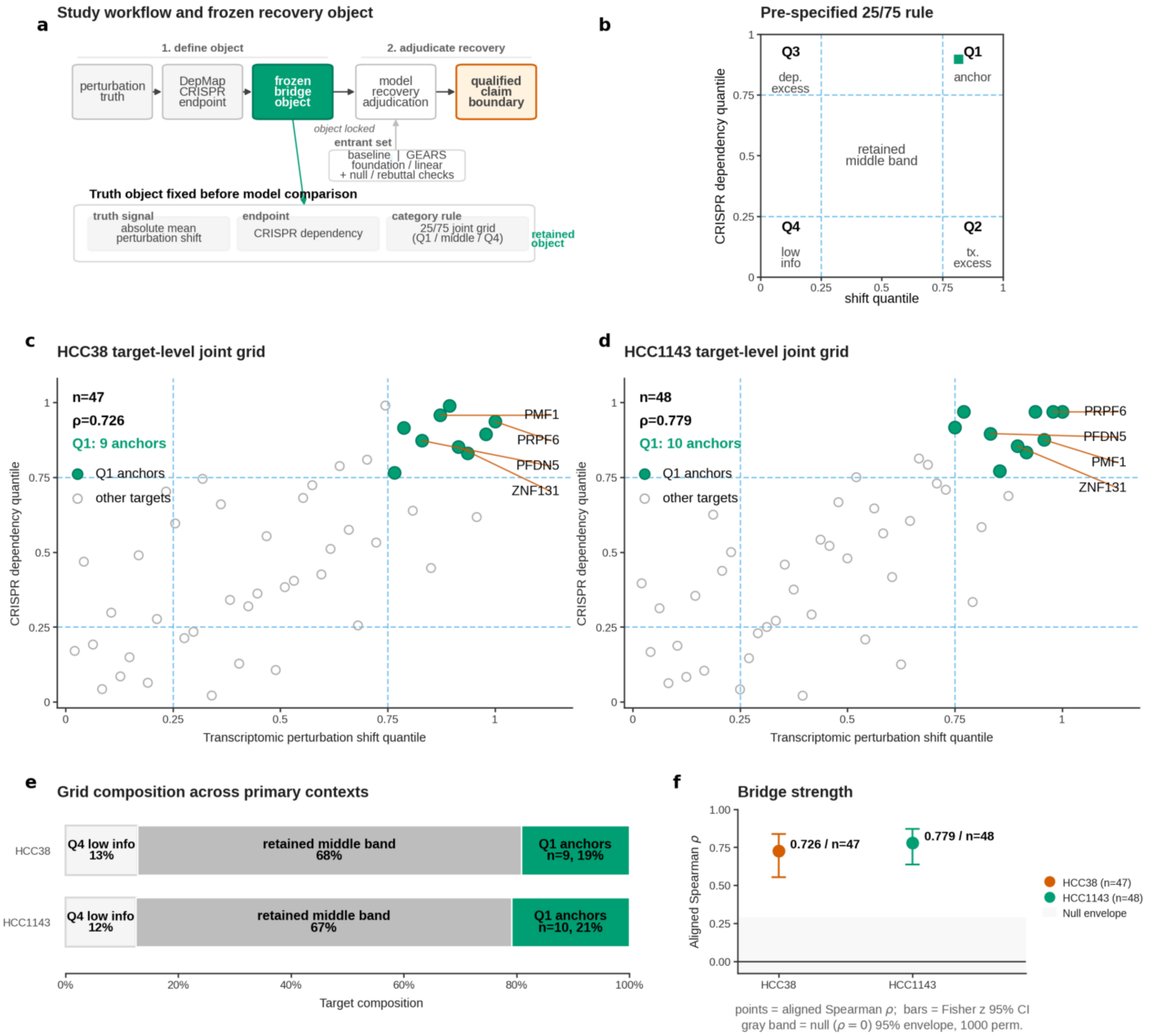


Fig. 1 | Framework element 1: A pre-specified perturbation-to-dependency bridge defines the truth object in HCC38 and HCC1143

a, Study workflow and frozen recovery object. The framework defines a phenotype-aligned truth object by aligning absolute mean perturbation shift (transcriptomic truth) to CRISPR DepMap dependency (fitness endpoint) before any model-scoring step. b, Pre-specified 25/75 category rule. Targets were classified on a pre-specified joint grid using the 25th and 75th percentiles of transcriptomic perturbation shift and CRISPR dependency, with corner-defined categories and a retained middle band. c, HCC38 target-level joint grid. Axes show within-context rank percentiles (shift_quantile and depmap_quantile), with n, aligned Spearman rho, and Q1 anchor count annotated in the panel. d, HCC1143 target-level joint grid. Axes show within-context rank percentiles, with n, aligned Spearman rho, and Q1 anchor count annotated in the panel. e, Grid composition across the two primary contexts. Q2 transcriptomic-excess and Q3 dependency-excess targets were not observed in either primary context and are retained as zero-count categories in the composition summary. f, Bridge strength summary.

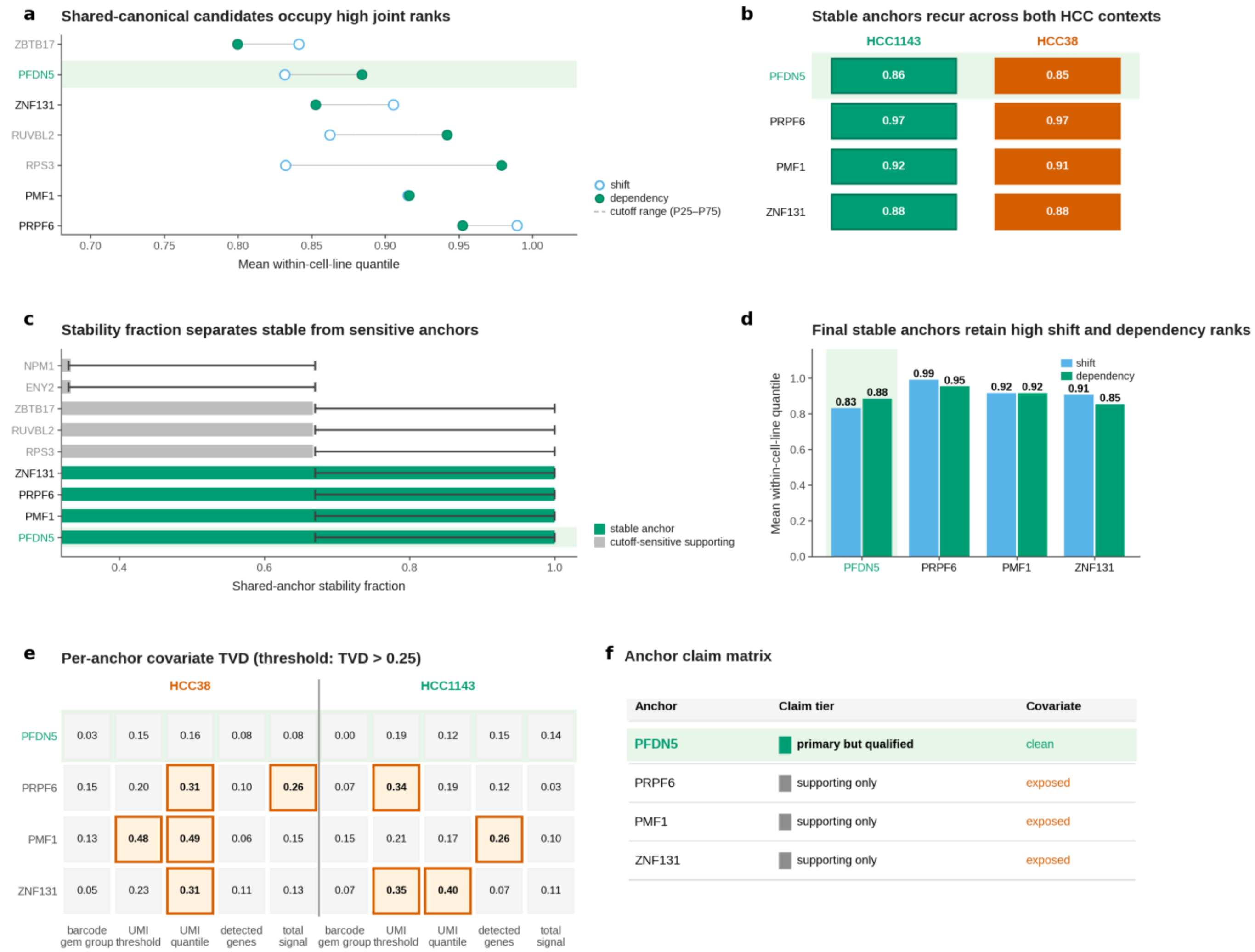


Fig. 2 | Framework element 2: Covariate-aware anchor tiering separates recurrent bridge support from target proof

a, Shared-canonical anchor ranking. Paired shift (open) and dependency (filled) within-cell-line quantile means for the seven shared-canonical candidates, ordered by the mean within-cell-line joint quantile (mean of shift and dependency quantile means). b, Low-ink recurrence matrix across the two primary HCC contexts (HCC1143 and HCC38) for the four final stable anchors. c, Stability fraction across stable and cutoff-sensitive shared-canonical objects. d, Final stable anchors retain high shift and dependency ranks across HCC38 and HCC1143, with numeric quantile means annotated per bar and the PFDN5 column highlighted; no covariate-level marking is drawn at this structural layer. e, Per-anchor covariate Total Variation Distance (TVD) matrix across five covariate axes (barcode gem group, UMI-threshold bin, UMI-quantile bin, detected-genes quantile bin, total-signal quantile bin) and the two HCC contexts, for the four final stable anchors. f, Compact anchor claim matrix (conclusion). PFDN5 is the only primary-but-qualified anchor and is covariate-clean; PMF1, PRPF6, and ZNF131 are retained as supporting only because panel e shows they are covariate-exposed despite being structurally stable.

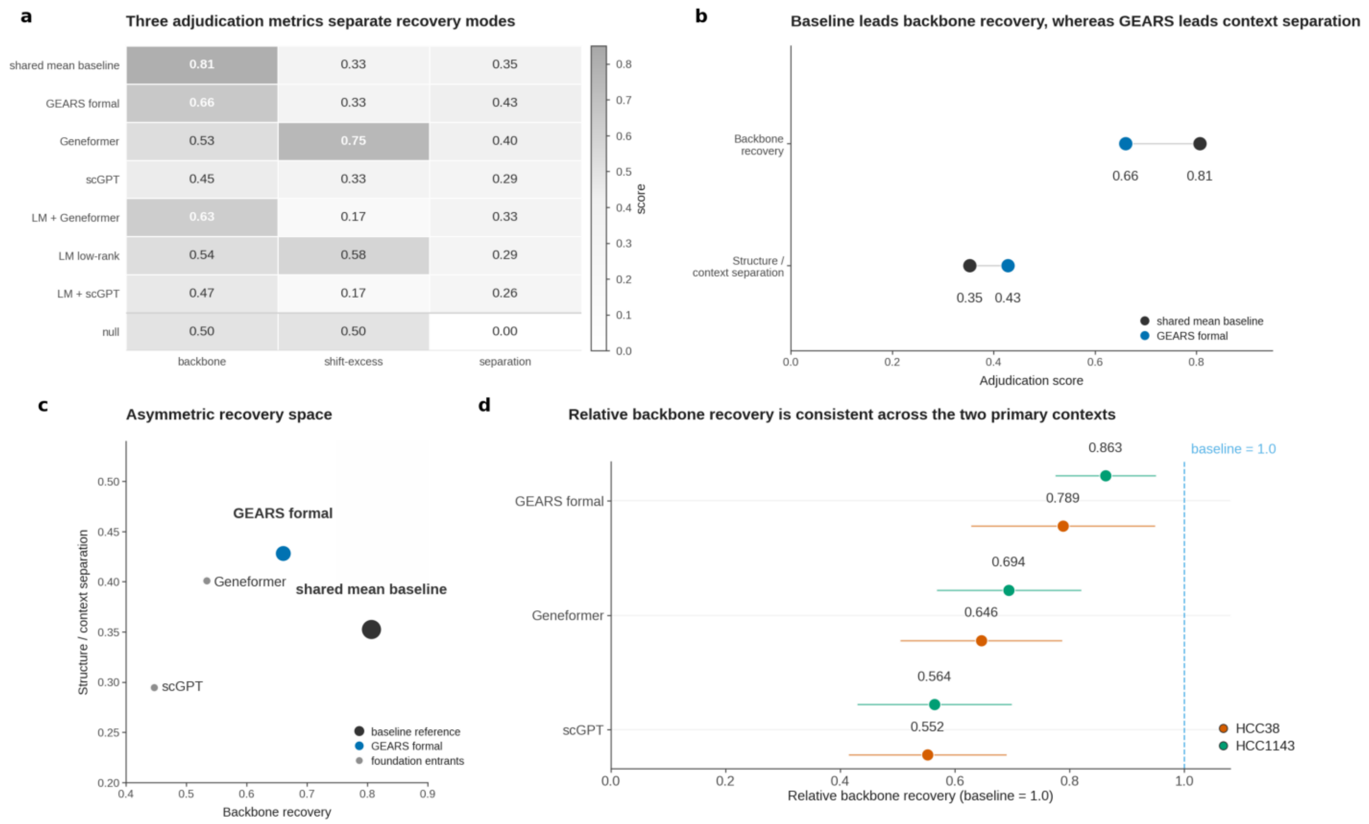


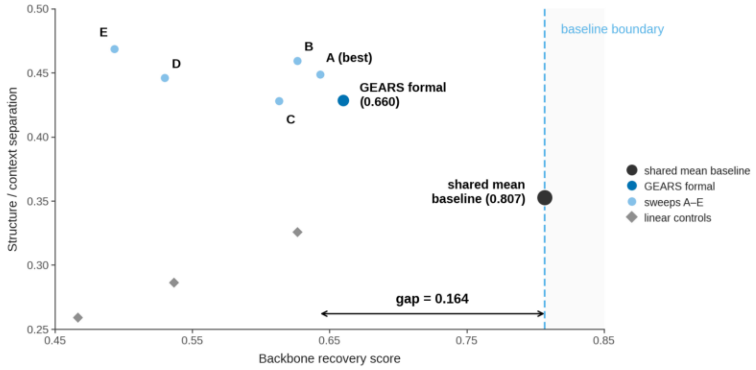
Fig. 3 | Framework element 3: Architecture-aware model adjudication reveals an asymmetric recovery pattern

a, Three adjudication metrics separate recovery modes across entrants. b, Baseline leads backbone recovery, whereas GEARS leads context separation. c, Entrants occupy an asymmetric recovery space. Backbone recovery versus structure-versus-context separation is plotted for the shared-mean baseline, the formal GEARS recipe, foundation-model entrants, and the linear controls.

a Pre-specified finite-budget GEARS candidates

Candidate	Epoch	LR	WD	Role
GEARS formal	30	1×10^{-3}	1×10^{-6}	reference recipe
A (best)	30	2×10^{-3}	1×10^{-6}	best sweep candidate
B	20	1×10^{-3}	1×10^{-6}	sweep candidate
C	30	1×10^{-3}	1×10^{-5}	sweep candidate
D	30	5×10^{-4}	1×10^{-6}	sweep candidate
E	40	1×10^{-3}	1×10^{-6}	sweep candidate

b Prespecified rebuttal candidates do not close the backbone gap



c No tested rebuttal candidate closes the backbone gap to the shared-mean baseline

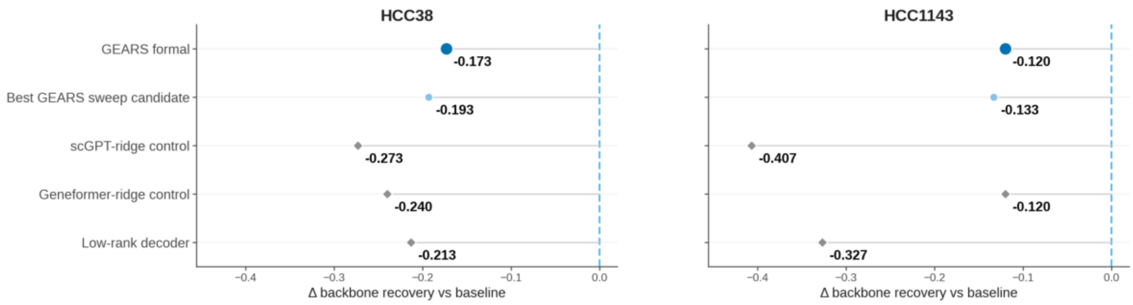


Fig. 4 | Framework element 4: Finite-budget rebuttal tests with a pre-specified stop rule do not close the backbone recovery gap

a, Pre-specified finite-budget GEARS candidates. Six pre-specified recipes were evaluated under an unchanged truth object and scoring system: the formal GEARS recipe and five one-parameter local variants varying epochs, learning rate, or weight decay. b, Pre-specified rebuttal recovery space. The shared-mean baseline, GEARS formal, the five GEARS sweep candidates, and three embedding-based linear controls are placed in the same backbone-recovery versus structure/context-separation space. Diamond markers denote the three linear controls: scGPT-ridge, Geneformer-ridge, and the low-rank decoder. c, Per-context residual backbone gap. GEARS formal, the best GEARS sweep candidate, and the three linear controls are plotted as backbone-recovery differences relative to the shared-mean baseline in HCC38 and HCC1143.

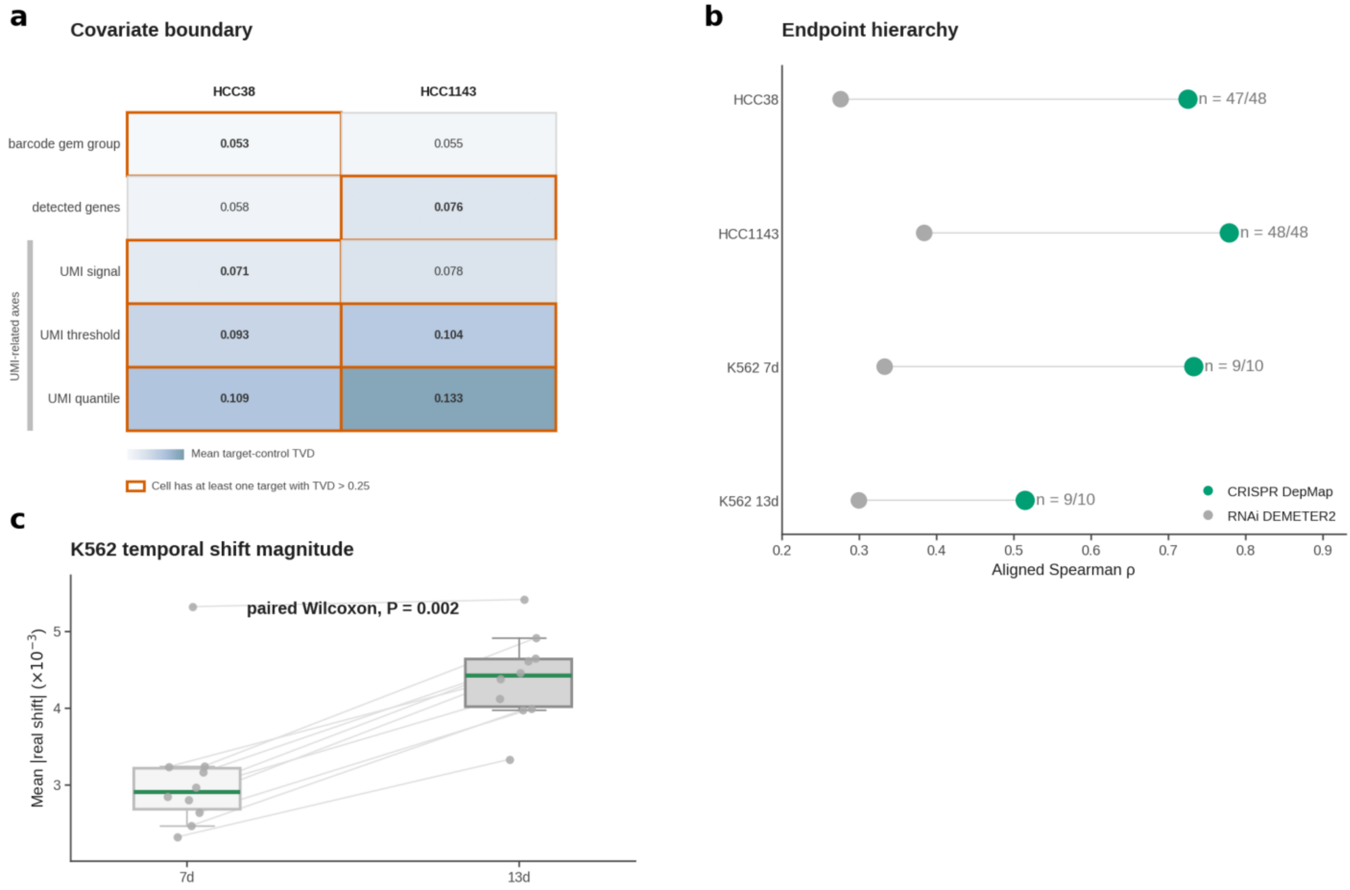


Fig. 5 | Framework elements 6 and 7: Temporal architecture, endpoint hierarchy, and covariate governance define the benchmark claim space

a, Covariate boundary in HCC38 and HCC1143 (Framework element 7). Mean target-control Total Variation Distance (TVD) is shown on a shared color scale for five covariate stratifications (barcode gem group, detected-gene bin, UMI signal, UMI threshold, UMI quantile). b, Endpoint hierarchy across all four bridge contexts (Framework element 6). c, K562 temporal stratification: stronger rank alignment at 7d despite smaller perturbation magnitude (Framework element 6). Paired boxplots summarize per-target mean absolute real shift at 7d and 13d; thin lines link paired targets; jittered points show the per-target values. d, A0/A1/B evidence-tier summary for K562 temporal evidence (Framework element 6).